

CABAUAU TOWER, Netherlands

WMO: 063480

Lat: 51.9703N

Lon: 4.9263E

Elev: 2

StdP: 101.30

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-4.1	-9.2	1.7	-4.2	-6.7	2.1	-3.3	15.1	9.1	13.5	9.2	2.8	70	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	28.1	20.5	26.0	19.5	24.2	18.4	21.4	26.6	20.3	24.7	19.3	23.0	3.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.3	23.8	18.6	13.5	22.3	17.8	12.7	21.2	62.0	26.7	58.2	24.9	54.8	23.0	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.3	9.0	-8.7	32.4	3.8	1.8	-11.4	33.7	-13.6	34.7	-15.7	35.7	-18.5	37.0		
(5)			-9.1	24.0	3.7	1.1	-11.8	24.8	-13.9	25.5	-16.0	26.1	-18.7	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	3.5	4.0	6.4	9.9	13.4	16.1	18.0	17.6	15.2	11.3	7.1	4.2
	DBStd	6.21	3.79	3.71	3.35	3.63	3.50	2.97	2.94	2.58	2.69	3.18	3.28	4.00
	HDD10.0	844	202	168	118	47	7	0	0	0	1	23	97	183
	HDD18.3	2901	460	400	370	254	157	79	42	44	101	218	337	439
	CDD10.0	1065	1	1	6	43	114	184	249	236	156	64	9	2
	CDD18.3	79	0	0	0	1	6	13	32	22	6	0	0	0
	CDH23.3	581	0	0	0	7	46	112	236	147	33	1	0	0
	CDH26.7	133	0	0	0	0	6	27	62	34	5	0	0	0
(14) Wind	WSAvg	4.2	5.2	5.0	4.6	4.1	4.1	3.8	3.7	3.4	3.5	4.2	4.4	5.0
(15) Precipitation	PrecAvg	864	71	67	54	45	61	71	89	87	76	77	85	83
	PrecMax	1078	131	141	96	103	132	148	189	210	200	185	144	162
	PrecMin	617	5	16	12	0	23	18	19	18	14	20	10	17
	PrecStd	120	35	30	28	26	29	37	43	50	42	36	32	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	13.8	17.7	24.3	27.2	30.0	31.5	30.3	26.7	21.8	16.2	13.2
		MCWB	11.1	10.2	12.7	15.9	19.3	21.3	21.5	21.2	20.0	17.4	14.1	11.8
	2%	DB	11.1	11.5	15.1	21.0	24.4	26.4	28.2	27.0	23.9	18.7	14.3	12.0
		MCWB	10.0	9.6	11.0	14.5	16.8	19.6	20.5	20.2	18.9	16.0	12.9	10.8
	5%	DB	9.9	10.4	13.0	18.3	22.3	23.9	25.9	24.6	21.7	17.1	13.0	10.8
		MCWB	8.8	8.8	10.3	13.0	16.0	18.4	19.8	19.1	17.6	15.0	11.7	9.7
	10%	DB	8.7	9.1	11.4	16.0	19.9	21.9	23.9	22.6	19.8	15.9	11.8	9.6
		MCWB	7.7	7.8	9.2	12.1	15.0	17.3	18.8	18.1	16.6	14.1	10.7	8.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.1	13.2	17.0	20.0	22.4	22.8	22.8	21.0	17.8	14.5	12.3
		MCDB	12.3	12.7	16.5	22.7	25.7	28.3	28.2	28.5	25.5	20.8	15.5	12.9
	2%	WB	10.1	10.0	11.7	15.0	18.1	20.3	21.5	20.9	19.4	16.4	13.1	11.0
		MCDB	10.9	11.2	14.2	20.0	22.4	25.1	26.9	25.6	23.0	18.4	14.3	11.9
	5%	WB	8.9	9.0	10.6	13.6	16.6	19.0	20.3	19.7	18.3	15.4	11.9	9.9
		MCDB	9.8	10.0	12.6	17.7	21.0	22.9	24.9	23.6	20.9	16.9	12.8	10.8
	10%	WB	7.9	8.0	9.5	12.4	15.3	17.8	19.2	18.7	17.2	14.4	10.7	8.7
		MCDB	8.7	9.1	11.3	15.5	19.2	21.3	22.8	21.7	19.4	15.7	11.7	9.6
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.6	7.3	9.0	8.9	8.9	8.5	8.7	8.1	6.7	5.3	4.5
		MCDBR	5.0	6.8	10.0	12.2	12.3	12.3	12.0	11.6	10.5	7.7	5.9	4.7
	5% WB	MCWBR	4.6	5.3	6.8	7.1	6.4	6.7	5.8	6.0	5.9	5.2	5.0	4.3
		MCDWR	5.0	6.1	8.9	11.4	11.0	11.1	10.9	10.5	9.4	7.4	5.8	4.7
(40) Clear-Sky Solar Irradiance	taub		0.344	0.366	0.401	0.426	0.433	0.427	0.437	0.427	0.402	0.380	0.356	0.340
		taud	2.293	2.269	2.185	2.160	2.194	2.228	2.210	2.255	2.311	2.332	2.308	2.282
	Ebn at Noon		636	733	777	807	821	829	812	798	772	709	613	569
		Edh at Noon	67	89	117	134	136	133	133	121	103	83	65	58
(44) All-Sky Solar Radiation	RadAvg		0.71	1.40	2.60	4.19	5.08	5.44	5.14	4.33	3.15	1.82	0.83	0.54
	RadStd		0.08	0.26	0.33	0.42	0.36	0.39	0.57	0.36	0.28	0.15	0.08	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page